



A fatal case of accidental asphyxia following nitrous oxide inhalation and alcohol consumption

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Abstract

Nitrous oxide (N₂O) abuse is becoming increasingly popular worldwide. Moreover, the use of N₂O combined with other substances, such as alcohol, is also common. Accidental deaths associated with N₂O abuse are rare in forensic practice, with most fatal cases involving continuous inhalation equipment or exposure in a confined space. In contrast, the inhalation of N₂O using balloons is traditionally regarded as a relatively safe method. In this report, we present an unusual fatal case of a 16-year-old girl who died after drinking alcohol and inhaling N₂O from balloons. The decedent was found in a prone position with the nose and mouth compressed against the bed. Cyanosis of nails, pulmonary and cerebral edema, and the positive expression of HIF-1 α in lung, myocardium and brain, were indicative of asphyxiation. Toxicological analysis revealed a peripheral blood ethanol concentration of 140 mg/dL and cardiac blood N₂O concentration of 74.5 μ L/mL. N₂O was also positively detected in the lungs, stomach contents, gastric air, and the brain. In conclusion, we determined that the decedent died from accidental asphyxia related to N₂O inhalation, prone positioning, and alcohol consumption. We also investigated chronic N₂O abuse from a forensic perspective. This report aims to help forensic pathologists manage similar cases. It also reminds N₂O abusers of potential dangers.

Keywords Nitrous oxide abuse · Alcohol · Accidental asphyxia

Introduction

Nitrous oxide (N₂O), also known as “laughing gas”, is an inhaled anesthetic that is medically widely used because of its rapid-onset anesthetic and analgesic properties [1]. It can inhibit excitatory glutamatergic neurotransmission via noncompetitive antagonism of N-methyl-D-aspartic acid (NMDA) receptors, activate kappa opioid receptors, and stimulate the release of endogenous opioids [1–3]. In recent

years, owing to its euphoric and hallucinogenic effects, recreational use has become increasingly popular, particularly among young people. N₂O is not strictly regulated in most countries, thus it can be easily obtained [1]. Abusers commonly inhale it from balloons, which is considered safe. The most common occasions of N₂O abuse are house parties and festivals, in which its concurrent use with other substances, such as alcohol, is prevalent [1, 4]. Prolonged N₂O abuse can cause functional or absolute vitamin B12 deficiency, which may result in hyperhomocysteinemia and demyelination in the nervous system, most commonly affecting the cervical spinal cord and peripheral nerves in the lower limbs, as well as megaloblastic anemia [3, 5]. N₂O inhalation may rarely lead to fatalities that often occur when using a bag or mask over the head, or when opening an N₂O tank in an enclosed space [1, 6]. In this report, we present a rare case of a young girl who died in the prone position after N₂O inhalation from balloons and alcohol consumption. This report aimed to help forensic pathologists address similar cases and raise public awareness of the dangers of N₂O abuse.

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Case presentation

Case history

A 16-year-old girl was found dead in the prone position on the bed. Her head was buried between two pillows (Fig. 1a). Dark-purple livor mortis was present on the ventral surface of the corpse and was especially pronounced in the head and facial regions. The livor mortis branched under pressure. Her nose and mouth were compressed, with no visible liver mortis in the surrounding area (Fig. 1b). Her nails (fingers/toes) exhibited mild cyanosis.

A balloon and mobile phone were discovered adjacent to the decedent's head (Fig. 1c). Two cylinders (common packaging used in the underground N₂O trading market in China), each connected to a short hose, were found on a bedroom balcony (Fig. 1d). The scene did not show any signs of violence. No other addictive drugs or poisons were identified. According to the investigation, the decedent left her hometown and began working independently after graduating from middle school. She was employed

as a hostess at KTV and lived with colleagues who had similar experiences. She had been abusing N₂O once a week for at least six months prior to her death. The exact amount used each time is unknown. The medical history was unremarkable, except for N₂O use and drinking. The evening preceding her death, she consumed alcohol with a client before returning home for dinner. Later, she and her roommate inhaled N₂O together. She was found dead at approximately 11 a.m. the next day.

Autopsy and histological examination

The corpse was frozen at -20°C until the autopsy was performed 11 days later. The decedent was 157.0 cm tall. During external examination, no remarkable injuries were found, including in the oral mucosa. Internal examination revealed significant congestion and edema in both lungs. The right lung weighed 510 g, whereas the left lung was not weighed as it was intended for toxicological analysis. The brain weighed 1,375 g and showed moderate edema. Histological examination revealed that most of the alveoli were filled with eosinophilic edematous fluid (Fig. S1a). Diffuse edema

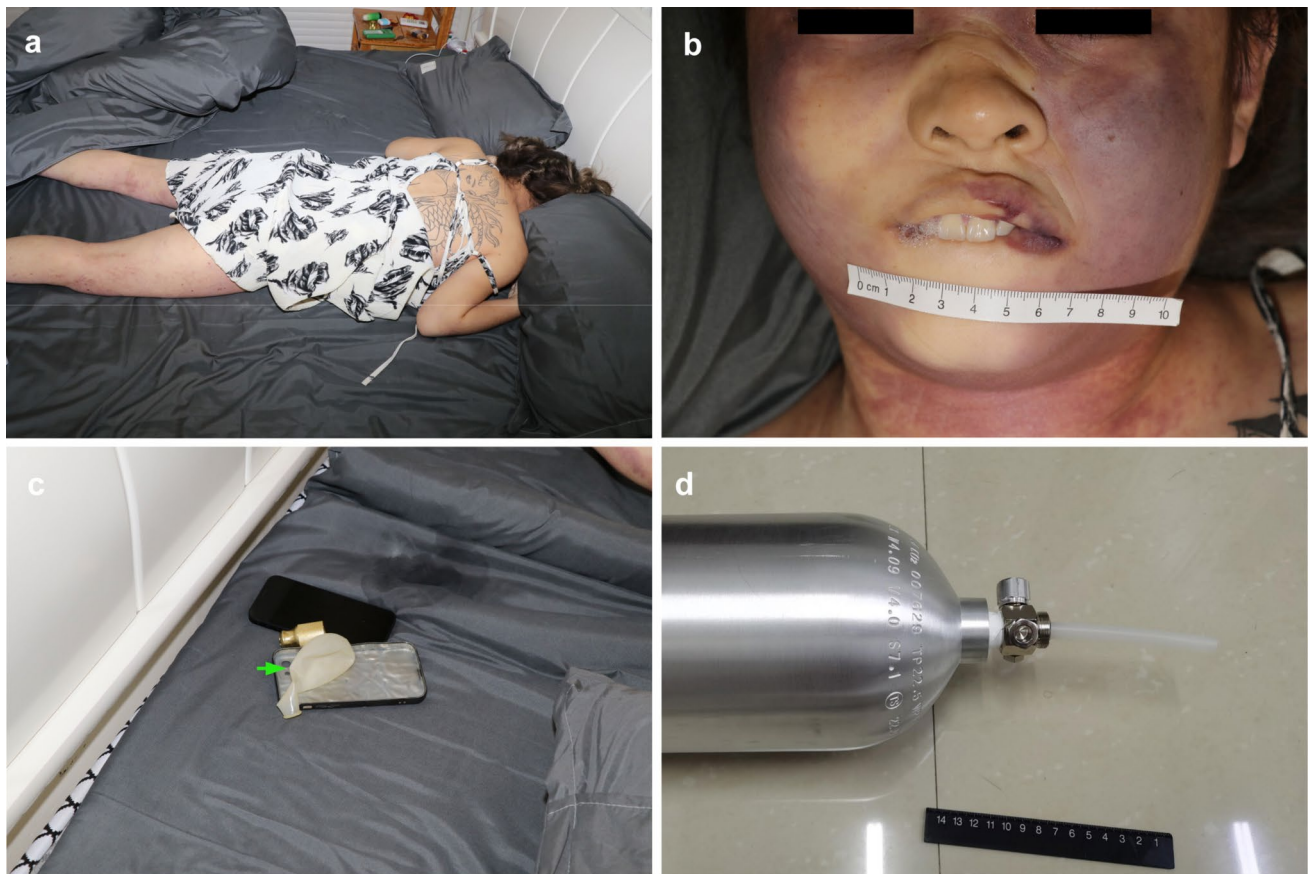


Fig. 1 **a** The decedent was found in a prone position. **b** The decedent's nose and mouth was compressed against the bed, and no livor mortis was found in the surrounding area. **c** The balloon was next to her head (green arrow). **d** The cylinder with a hose was found on the balcony

was observed in the brain (Fig. S1b). Sporadic vacuolation of the myelin sheath was observed in the lateral columns of the upper cervical spinal cord (Fig. S1c). No significant morphological abnormalities were observed in the vagus or sciatic nerves. Luxol fast blue (LFB) staining revealed no significant demyelination (patchy pale staining) in the nervous system, including in the brain, brainstem, and cervical spinal cord. Immunohistochemistry testing revealed positive expression of HIF-1 α , a potential biomarker for asphyxia death, in the lungs, myocardium, and brain (Fig. S2a–c). No remarkable findings were found in the other organs, except for congestion.

Toxicological and other findings

Peripheral blood samples were collected from the decedent at the scene and subjected to common toxicological analysis two days after the incident. During the autopsy, cardiac blood, lung, stomach contents, gastric air, and part of the brain were sealed and stored in headspace vials at -20°C for N_2O analysis. Some cardiac blood was collected in EDTA tubes at 4°C and was immediately sent to the hospital for vitamin B12 and homocysteine testing.

N_2O was identified and quantified using HS-GC/MS two months later. Samples of 1.0 mL blood, 100 μL of gas, 1.0 g of solid sample were put into 20 mL headspace sample vials separately, and sealed. The headspace sample vials were incubated in the automatic headspace sampler (Thermo Fisher Triplus 300) and at 65°C for 15 min, then 1 μL of headspace gas in the vials was sampled and measured by Thermo Fisher TRACE 1300-ISQ QD 300 GC–MS system. An Agilent GS-CARBONPLOT quartz capillary column (30 m $\times$ 0.32 mm $\times$ 3.00 μm) was used for chromatographic separation. Ultrahigh pure helium (He) was used as the carrier gas with a flow rate of 1.0 mL/min. The temperature program started at 35°C and held for 5 min, followed by an increase to 140°C with $20^{\circ}\text{C}/\text{min}$ temperature gradient, and held for 2 min. The mass spectrometer was operated in electron ionization (EI) mode at 70 eV, with the ion source temperature set at 230°C . Full-scan mass spectra were set in the range of 10–200 amu with no solvent delay applied (0 min). The qualitative ions were m/z 44, m/z 30 and m/z 28, precursor/product ion pairs were confirmative ions, and the quantification ion was m/z 30. The N_2O retention time was 2.04 min. Quantification was performed using an external standard calibration curve. To create the standard curve, 1 μL , 5 μL , 10 μL , 20 μL , 50 μL , 100 μL , and 200 μL of N_2O standard gas were added to blank blood samples. Each concentration point was measured in duplicate. The standard curve equation was $Y = 45928X + 36543$, $R^2 = 0.997$.

The limits of detection (LODs) and of N_2O in different samples were determined by a signal-to-noise approach. For blood samples, the LOD was 0.5 $\mu\text{L}/\text{mL}$, for gas samples, the

LOD was 0.05 $\mu\text{L}/\text{mL}$, and for solid samples, the LOD was 0.5 $\mu\text{L}/\text{g}$. The limit of qualification (LOQ) for blood samples was 0.5 $\mu\text{L}/\text{mL}$. Precision and accuracy were evaluated by intra-day and inter-day parameters in blood as sample matrix. Intra-day and inter-day precision was 12.4–18.4% (the samples were analyzed over 5 days). And the accuracy results were all more than 96%.

Ultimately, N_2O was positively detected in the cylinders, the decedent's cardiac blood, lung, stomach contents, gastric air, and brain, with an N_2O concentration of 74.5 $\mu\text{L}/\text{mL}$ in the cardiac blood (Fig. 2). Routine toxicological analysis of peripheral blood revealed an ethanol level of 140 mg/dL, while illicit drugs (morphine, methamphetamine, and ketamine), benzodiazepines (diazepam and eszopiclone), and common lethal poisons (strychnine and organophosphorus pesticides) tested negative. The vitamin B12 level in the cardiac blood was $> 1,526$ pg/mL, with a reference range of 180–914 pg/mL. The homocysteine level was 25.4 $\mu\text{mol}/\text{L}$, with a reference range of 6–17 $\mu\text{mol}/\text{L}$. DNA profiles from the balloon neck and cylindrical hose matched the blood samples from the decedent.

Discussion

Comprehensive autopsy and histological examination ruled out fatal diseases and injuries as the causes of death. Non-specific signs of asphyxia, including nail cyanosis, pulmonary edema and cerebral edema, were observed. HIF-1 α is a transcription factor expressed in response to hypoxia, which was recommended by some forensic pathologists to assist the diagnosis of asphyxia cases [7]. And it positively expressed in the lung, myocardium, and brain of the decedent, supporting the diagnosis of asphyxia. Based on autopsy findings, investigations, and toxicological analysis, the cause of death was ultimately determined to be accidental asphyxia associated with N_2O inhalation, the prone position, and alcohol consumption.

The mechanisms of death caused by N_2O are deemed to be hypoxemia and asphyxia due to the displacement of oxygen and diffusion hypoxia [1]. Research has shown that N_2O does not cause major respiratory depression but can depress the sensation of dyspnea [8]. Such deaths often result from abusers failing to stop the continuous inhalation devices timely. However, the decedent in this case inhaled N_2O from balloons in a spacious bedroom, which is unlikely to cause long-term hypoxia. Therefore, the prone position was essential for the decedent's death. This abnormal position may obstruct the nose and mouth. This has also been observed in some cases of accidental asphyxia in infants [9] and substance abusers [10]. Additionally, alcohol consumption also contributed to the death of the deceased. Although the specific interaction mechanisms between N_2O and ethanol are

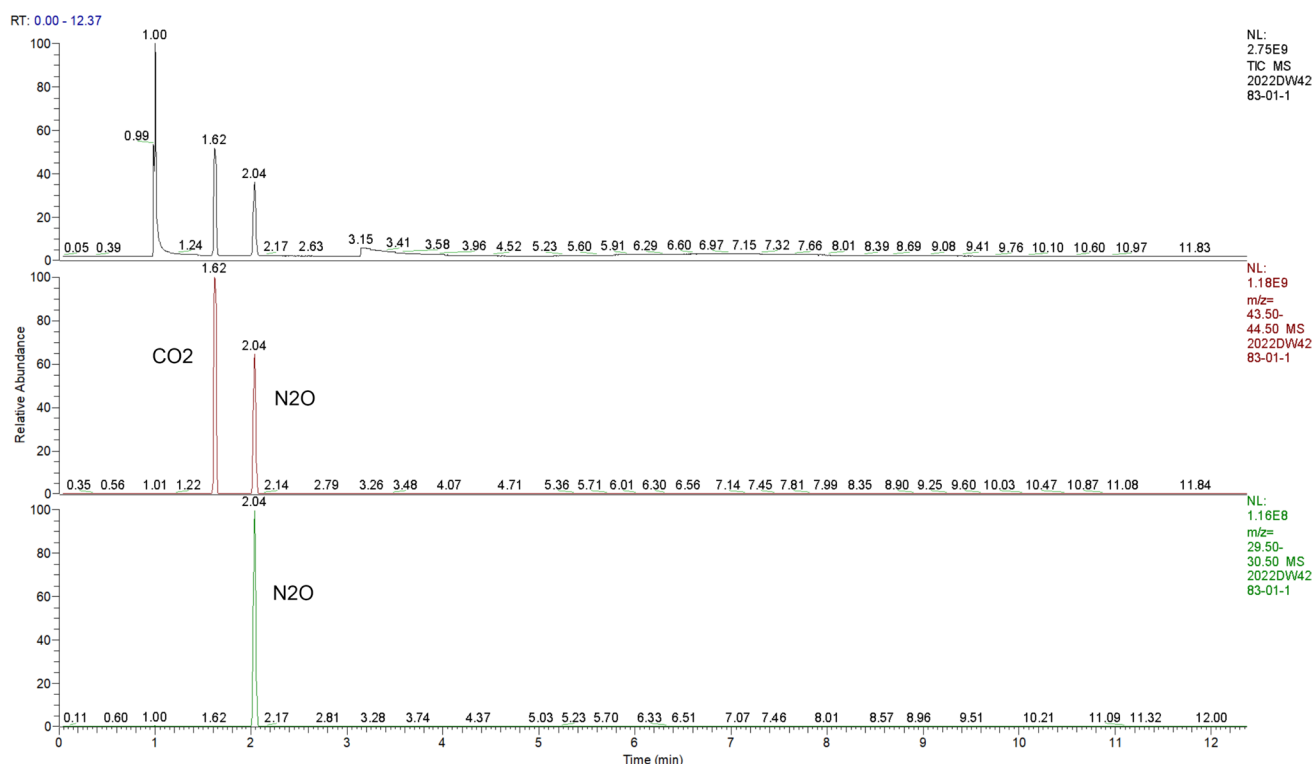


Fig. 2 Chromatograms of the decedent's blood samples showing retention time and separation of nitrous oxide and carbon dioxide

not fully understood, studies show that people tend to consume more N_2O after drinking [11]. In addition, both N_2O and alcohol can inhibit the NMDA subtype of glutamate receptors, potentially exacerbating central nervous system depression [12, 13].

This rare case also exhibited some common characteristics with other N_2O -caused asphyxia deaths reported in the literature. Such deaths are usually accidental [14–17]. In addition to abuse, medical errors may also be possible causes. And a small number of these deaths were suicidal [18]. Similar to the presented case, autopsy findings and routine histological examinations in N_2O -caused deaths typically reveal nonspecific signs of asphyxia, such as cyanosis [19], visceral congestion [8, 20], pulmonary edema [21], cerebral edema [22], and occasionally conjunctival petechiae [23].

Toxicological analysis of these cases is crucial. N_2O is commonly analysed using HS-GC–MS. The half-life of N_2O is very short (a few minutes) and after inhalation, most N_2O is rapidly eliminated through the lungs [2]. Thus, N_2O can only be detected when death occurs soon after inhalation [2]. Blood, urine, liver, bile, kidneys, adipose tissue, brain and air samples from the stomach and trachea have been used for toxicological analysis of N_2O -caused fatalities [16]. N_2O is rarely produced postmortem, but may be lost from biological samples during storage and detection, therefore, it has only

been qualitatively detected in a number of N_2O inhalation cases [16]. A strict quantitative analysis of N_2O can only be conducted in blood, urine, and gas samples because of the heterogeneity of solid samples [16]. Dimaio et al. reported N_2O blood concentrations ranging from 4.6 to 8.97 mL/dL and in the lung air ranging from 12.3% to 22.7% [20]. Poli et al. reported blood concentrations of 11.29–2,152.04 mg/L, urine concentrations of 95.11 mg/L, and lung, stomach, and tracheal air concentrations ranging from 5.28 to 83.63 g/m³ [16]. However, to date no lethal blood N_2O concentrations have been recognized. In this case, the quantitative analysis of N_2O in cardiac blood and the qualitative analysis in the lung, brain, gastric air, and stomach contents support its unignorable role in the asphyxia death of the decedent.

We also investigated the deceased's long-term abuse of N_2O history from a forensic perspective. Clinically, N_2O abusers with neurological symptoms often exhibit decreased serum vitamin B12 and elevated homocysteine levels, with homocysteine being a more sensitive indicator [24]. Morphological findings in chronic N_2O abuse include spongy degeneration and vacuolation of the myelin sheath in the long spinal myelinated tracts, as well as scattered axonal swelling [22]. A sural biopsy study revealed decreased density of myelinated nerve fibers, increased density of thinly myelinated fibers, and axonal regeneration [25]. The frequency and duration of abuse are related to such changes

but vary among individuals [26]. In this case, despite the decedent's elevated levels of peripheral blood vitamin B12 and no significant demyelination in the nervous system, slightly higher homocysteine levels and sporadic vacuolation of the myelin sheath were observed, which was deemed to be associated with N₂O abuse. The abnormally high vitamin B12 levels in the decedent may have resulted from prior supplementation. Some Chinese retailers sell vitamin B12 in combination with N₂O [24]. It is worth noting that these changes reflect long-term N₂O abuse, and their forensic use may be limited, as vitamin B12 declines and homocysteine rises in stored blood over time [27, 28].

Conclusion

There have been no explicit legal provisions for controlling N₂O as a psychoactive substance in many countries, including China, which places it in the gray area of substance abuse and warrants greater attention from forensic pathologists. A comprehensive autopsy and histological examination are necessary to exclude other causes of death. The toxicological analysis of N₂O in biological samples is essential for forensic identification. Immunohistochemical examination may help support the presence of asphyxia. Measurements of vitamin B12 and homocysteine levels in the blood as well as histological examination of the nervous system may further indicate a history of N₂O abuse. We strongly recommend more regulation of these grey-area substances and remind people to be aware of the danger of N₂O abuse, especially when mixed with alcohol. We also call for greater care and proper guidance for the vulnerable population.

Key points

1. Fatalities caused by inhaling N₂O from balloons are seldom reported.
2. N₂O was detected in the decedent's cardiac blood, lung, stomach contents, gastric air, and brain, providing a valuable reference for selecting toxicological samples in similar cases.
3. N₂O requires more regulation, and better social care is essential for preventing substance abuse in vulnerable population.

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Author contributions All authors contributed to the study conception and design. Autopsy, histopathological examination were performed by Huine Liu, Yiwu Zhou and Yijie Duan. The resources preparation was performed by Yi Yang. The toxicology analysis was performed by Man Liang and Jinghong Ma. The first draft of the manuscript was written by Huine Liu and Man Liang. The manuscript was reviewed and edited by Hongmei Dong and Yijie Duan. All authors read and approved the final manuscript.

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Declarations

Ethics approval and informed consent This article does not contain any studies involving human participants or animals performed by the author. Informed consent was obtained from the deceased's father in the study.

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